A Robust Multispectral Point Cloud Generation Method Based on 3-D Reconstruction From Multispectral Images

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***Abstract*— Multispectral point cloud is a novel type of data rich in spectral and spatial information. The 3-D reconstruction is a low-cost solution for acquiring multispectral point cloud. However, most of the existing methods have been developed for RGB images, which are inapplicable to multispectral images due to the special structure of multispectral sensors and the nonlinear intensity differences. In this article, a robust 3-D reconstruction method for multispectral images is proposed to generate multispectral point cloud by harnessing their spatial and spectral information. Considering the characteristics of multispectral image acquisition, reflectance correction and band alignment steps are introduced into the proposed method, aiming to reduce the impact of band differences and spatial errors on 3-D reconstruction. Subsequently, a fused multispectral feature extraction is employed to provide more potential reconstruction feature points. To reduce the mismatched feature points induced by the spectra of vegetation regions, a normalized digital vegetation index (NDVI)-guided feature matching algorithm is proposed that provides accurate correspondence calculation for multispectral image reconstruction. The experiments compared with several well-known methods and commercial software on two datasets have shown superior reconstruction performance.**

***Index Terms*— 3-D reconstruction, multispectral feature extraction, multispectral images, multispectral point cloud.**

I. Introduction

MULTISPECTRAL point cloud, which combines the complementary advantages of multispectral image (MSI) and point cloud in spatial and spectral, has emerged as a novel data modality in the field of remote sensing [1], [2], [3]. This new type of data allows methods applied to 2-D multispectral image to be extended to 3-D point cloud, with the potential to improve the efficiency and accuracy of multiple application tasks, such as semantic segmentation and target detection [4], [5], [6], [7]. Multispectral point cloud contains rich spectral information of surface features [8], [9], [10], [11], [12] and the 3-D spatial characterization capability [13], [14], which can enhance the representation of the observation scene with greater completeness and accuracy [15], [16], [17]. The methods of obtaining multispectral point cloud data in current research can be divided into the following three types, including development of multispectral LiDAR, fusion of multispectral image and LiDAR point cloud, and image-based 3-D reconstruction.

Traditional LiDAR systems typically detect at a single wavelength due to technological of laser scanners. To broaden the application of LiDAR in vegetation and water areas, certain scholars initiated the development of dual-band and multiband LiDAR systems. Rall and Knox [18] developed a LiDAR system equipped with two semiconductor laser diodes operating at 660 and 780 nm, which is conducive to the observation of vegetation rich in chlorophyll. The renowned LiDAR company Riegl developed a dual-band LiDAR system specifically designed for measuring water depth [19]. Titan introduced the pioneering commercially available multispectral LiDAR system, encompassing bands at 550, 1064, and 1550 nm [20]. Furthermore, the invention of supercontinuum laser source technology has enabled the expansion of LiDAR's spectral bandwidth [21]. Several researchers have developed multispectral or hyperspectral LiDAR systems based on this approach [22], [23], [24]. However, most of these systems remain in the experimental phase, lacking fully matured products.

Moreover, the fusion of active detection LiDAR and passive imaging sensors has emerged as a current research focus [25], [26], [27]. The key challenge lies in minimizing the registration error between the 3-D point cloud and the 2-D image. A direct approach involves point cloud filtering, generating a digital surface model with height information, and fusing it with the image. However, this method sacrifices the 3-D structural information within the point cloud. Certain studies calibrate sensor parameters to minimize the distance between conjugate feature points, thus reducing the spatial disparity between the data [28], [29]. Building upon this foundation, Gu et al. [30] and Brell et al. [31] employed ray tracing to project and register the point cloud and hyperspectral data, thereby further mitigating the registration error resulting from variations in sensor detection principles. In the realm of fusion detection system design, Gu et al. [32] integrated separate multispectral and LiDAR sensors onto a UAV system, enabling the generation of a multispectral point cloud through simultaneous multisensor data collection.

Due to advancements in camera resolution and data processing technology within the field of computer vision,

Manuscript received 21 July 2023; revised 9 October 2023; accepted 17 October 2023. Date of publication 20 October 2023; date of current version 1 November 2023. This work was supported by the National Science Fund for Outstanding Young Scholars under Grant 62025107. *(Corresponding author: Xian Li.)*

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Digital Object Identifier 10.1109/TGRS.2023.3326153

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an image-based 3-D reconstruction method has emerged as a cost-effective solution and has gained widespread adoption [33]. This technique aims to extract the 3-D structure and spatial coordinates of an observed scene from a set of overlapping 2-D images. The specific steps involve: 1) extracting key features from the images and establishing image matching pairs by comparing them with neighboring images; 2) utilizing the information from the matching pairs to generate reconstructed feature tracks and iteratively computing the pose information of the reconstructed images; and 3) recovering the 3-D spatial information of the observed scene by leveraging the pose information and the principle of parallax, where cameras at varying positions produce distinct perspective effects on a stationary object [34], [35]. The advancement of structure from motion (SFM) and multiview geometry (MVS) technology has further enhanced the flexibility of 3-D reconstruction, enabling its application in scenarios without ground control points. SFM automatically calculates camera parameters and relative pose relationships from the image set using feature extraction and iteratively optimizes them to generate sparse point cloud [36], [37]. MVS utilizes the results of sparse reconstruction to calculate the depth map of the corrected image, thereby generating dense point cloud [38], [39]. Several commercial software programs, such as Agisoft and Pix4D, have been developed and demonstrated good performance [40].

In conclusion, image-based 3-D reconstruction technology is the most convenient, flexible, and cost-effective method for acquiring point cloud. However, current 3-D reconstruction methods are limited to grayscale or RGB images. Multispectral images contain a higher number of spectral bands, which introduces three primary challenges as outlined in the following. First, multispectral cameras typically employ a multilens design, with each lens equipped with a specific filter as in Fig. 1. Images from different bands exhibit relative positional differences, and the number of images is directly proportional to the number of bands, resulting in increased computational cost for 3-D reconstruction. Second, nonlinear intensity differences exist between bands, rendering existing feature extraction algorithms, such as SIFT [41], unsuitable for extracting features from multispectral images due to their heavy reliance on intensity relationships. Third, distinct ground objects exhibit varying band reflection characteristics. Incorrect matching relationships during the feature matching process can result in noticeable distortions in the reconstruction results [42].

To tackle these challenges, a multispectral 3-D reconstruction method is proposed in this article, which exhibits efficient and robust performance in multispectral point cloud generation. The first step involves calibrating the radiance of multispectral images to reduce the amplification of nonlinear intensity differences. Subsequently, the camera parameters are employed to spatially align the multiband images, mitigating spatial errors and enhancing reconstruction efficiency. Using the preprocessed multispectral image as input, the proposed pipeline utilizes the fused multispectral feature extraction algorithm and normalized digital vegetation index (NDVI)-guided feature matching to achieve accurate correspondence of multispectral images. Finally, SFM is employed to reconstruct

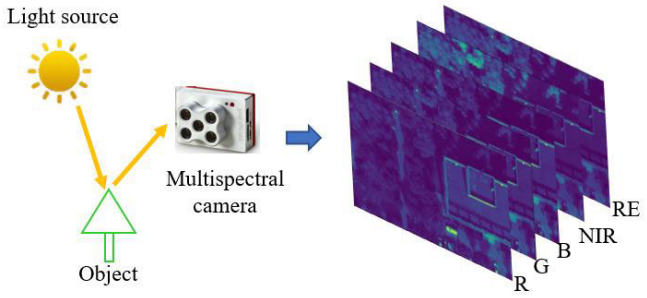


Fig. 1. Multispectral imaging system and imaging results.

the relative spatial relationships among the images and compute their depth maps, which are then used to generate multispectral point cloud. To assess the effectiveness of the proposed method, we conducted experiments using two UAV multispectral image datasets. The experimental results demonstrate a significant improvement in the reconstruction quality of multispectral images. The primary contributions of this study can be summarized as follows.

1) A workflow for 3-D reconstruction of multispectral images is proposed for generating multispectral point cloud. The method introduces reflectance correction and band alignment steps in 3-D reconstruction for the first time, which improves the efficiency of 3-D reconstruction and reduces the alignment error between bands.
2) In order to cope with the challenges of nonlinear intensity differences in multispectral images, a fused multispectral feature extraction algorithm is proposed, which uses reflectance calibrated multispectral images, retains the robust features in each band of the multispectral images, and removes redundant points. The spectral information is fully utilized and the number of candidate reconstruction feature points is increased to improve the effect of multispectral 3-D reconstruction.
3) An NDVI-guided matching method is proposed to address the problem of feature mismatching caused by spectral differences in vegetation regions. The method leverages the characteristics of vegetation regions in multispectral images to significantly reduce the cases of mismatching and improve the robustness and accuracy of 3-D reconstruction results.

The rest of this article is organized as follows. Section II introduces the method of 3-D reconstruction of multispectral images in detail. Section III describes the experimental results and analysis of the proposed method on two datasets. Finally, the conclusions are drawn in Section IV.

II. 3-D Reconstruction of Multispectral Images

The flowchart of the 3-D reconstruction of multispectral images is shown in Fig. 2, where the input is a multiband multispectral image set, and the output is a densely reconstructed multispectral point cloud. The proposed method is mainly divided into the following three steps: image preprocessing, correspondence computation, and 3-D information recovery, which will be described in detail next.

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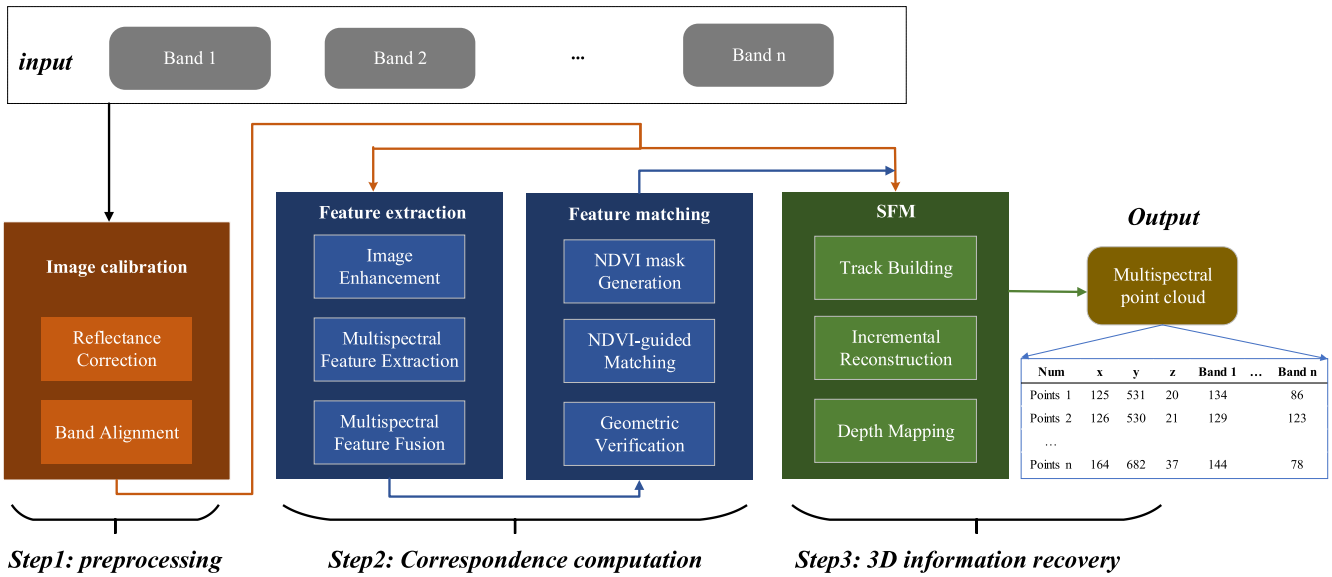


Fig. 2. Flowchart of the proposed 3-D reconstruction for UAV multispectral imagery.

A. Preprocessing of Multispectral Images

1) Reflectance Correction of Multispectral Images: The intensity values of multispectral images collected by UAVs are affected by environmental changes, such as lighting conditions and shooting angles, posing challenges for 3-D reconstruction based on image correlation. To mitigate the impact of other uncertain factors, researchers commonly employ the reflectance of an object to characterize its spectral properties. Consequently, it is necessary to apply a multispectral radiometric correction model that restores the digital number (DN) of the multispectral image to radiance, effectively eliminating the influence of sensor noise. Then, the radiance is converted into reflectance with the assistance of a downwelling light sensor and a standard reflectance panel. The specific correction model is given as follows:

$$L = g \cdot \mathbf{V}(x, y) \cdot \mathbf{R}(y) \tag{1}$$

where g is the sensor gain factor, $\mathbf{V}(x, y)$ represents the radial vignette model, and $\mathbf{R}(y)$ denotes the row gradient correction model. The radiometric correction process employs a radial vignette model to address the nonuniformities in the optical properties of the lens, which result in darker edges on the imaging plane compared to the center. The correction scaling factor for each pixel intensity can be calculated by (1)

$$\mathbf{V}(x, y) = \frac{1}{\mathbf{k}} \tag{2}$$

$$\mathbf{k} = 1 + \mathrm{k}_0 r + \mathrm{k}_1 r^2 + \mathrm{k}_2 r^3 + \mathrm{k}_3 r^4 + \mathrm{k}_4 r^5 + \mathrm{k}_5 r^6 \tag{3}$$

$$r = \sqrt{(x - \mathrm{c_x})^2 + \left(y - \mathrm{c_y}\right)^2} \tag{4}$$

where $(\mathrm{c_x}, \mathrm{c_y})$ is the vignette center of the image, r is the distance of the pixel (x, y) from the vignette center, $\mathbf{k}$ represents the correction factor, and $\mathrm{k}_i (i = 0, 1, \ldots, 5)$ are six polynomial coefficients from the image metadata.

In addition, due to the limitations of parasitic light and MOS transistors, the radiation error of CMOS imager pixels needs to be considered. The correction model is given as follows:

$$\mathbf{R}(y) = \frac{I(x, y) - \mathrm{I_{BL}}}{t_e + \mathrm{a}_1 y - \mathrm{a}_2 t_e y} \tag{5}$$

where t_e is the image exposure time, $I(x, y)$ is the normalized raw pixel value at (x, y), $\mathrm{I_{BL}}$ is the normalized black level value, and $(\mathrm{a}_1, \mathrm{a}_2)$ are the radiometric correction coefficients.

Finally, the reflectance data are generated by compensating for variations in illumination through the utilization of a standard reflectance panel and a downwelling light sensor. The radiometric correction model and reflectance panel are shown in Fig. 3.

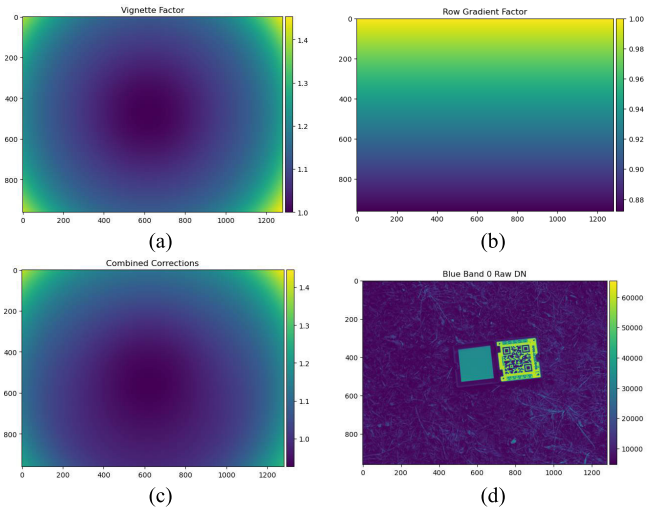


Fig. 3. Radiometric correction model and standard reflectance panel intensity image. (a) Radial vignette model, (b) row gradient correction model, (c) combined correction model, and (d) DN value of standard reflectance panel in blue band.

2) Band Alignment: Due to the physical separation of lenses in the multispectral sensor, there is a spatial misalignment among images captured in different spectral bands. Directly overlaying the spectral bands onto the images results in a

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spectral misalignment, causing a spatial–spectral error known as ghosting artifacts. In order to minimize the errors in the multispectral reconstruction results, the multispectral images were registered and utilized as input data for the process of 3-D reconstruction.

Geometric registration is an image processing technique used to align different images of the same scene, serving as an essential accuracy assurance in the field of remote sensing image processing [43]. Many studies have achieved good results in image registration [44], [45]. However, considering the computational burden of large data in 3-D reconstruction, this section adopts a band alignment method based on the multispectral camera structure. The process of aligning the bands of images consists of three steps: correcting distortion in each band image, establishing a registration transformation matrix between lenses of different bands, and cropping and merging the aligned images. The correction distortion is mainly to remove the radial distortion caused by the irregular change of the radial curvature of the camera lens

$$x' = x\left(1 + \mathrm{j}_1 r^2 + \mathrm{j}_2 r^4 + \mathrm{j}_3 r^6\right) \tag{6}$$

$$y' = y\left(1 + \mathrm{j}_1 r^2 + \mathrm{j}_2 r^4 + \mathrm{j}_3 r^6\right) \tag{7}$$

where (x', y') represents the corrected pixel coordinates and $\mathrm{j}_i (i = 1, 2, 3)$ are the compensation coefficients.

The registration transformation matrix is calculated using the Euler angles and offsets between the lenses of the camera. The calculation relationship between the rotation matrix of each band and the Euler angles is defined as follows:

$$\mathbf{R}(\varphi, \omega, \kappa) = \mathbf{R}(\varphi) \cdot \mathbf{R}(\omega) \cdot \mathbf{R}(\kappa) \tag{8}$$

where $(\varphi, \omega, \kappa)$ is obtained from the sensor parameters

$$\mathbf{R}(\omega) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\omega & -\sin\omega \\ 0 & \sin\omega & \cos\omega \end{bmatrix} \tag{9}$$

$$\mathbf{R}(\varphi) = \begin{bmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{bmatrix} \tag{10}$$

$$\mathbf{R}(\kappa) = \begin{bmatrix} \cos\kappa & -\sin\kappa & 0 \\ \sin\kappa & \cos\kappa & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{11}$$

Finally, the registered images need to be cropped to preserve their overlapping areas before merging them to produce a multispectral image.

B. Image Correspondence Computation

Feature extraction and matching are important steps in 3-D reconstruction. These steps involve detecting feature points in images and establishing correspondences between feature points in overlapping images. The information is then utilized to establish the projection relationship through geometric verification.

1) Feature Extraction: Due to the different conditions of image acquisition in the image set, images exhibit variations in resolution, intensity, and angle. To ensure unique identification of features across multiple images, it is essential for the features to exhibit invariance to both radiation and geometric changes. A feature extraction method with enhanced robustness is typically required for 3-D reconstruction. In particular, SIFT is extensively employed during the feature extraction stage of 3-D reconstruction due to its notable properties of position, scale, and rotation invariance. The construction of a Gaussian pyramid is employed to create the scale space, enabling the extraction of scale-invariant features

$$\mathbf{L}(x, y, \sigma) = \mathbf{G}(x, y, \sigma) * \mathbf{I}(x, y) \tag{12}$$

$$\mathbf{G}(x, y, \sigma) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{13}$$

where $\mathbf{L}(x, y, \sigma)$ is the result of convolving the image $I(x, y)$ with Gaussian blur $\mathbf{G}(x, y, \sigma)$ at scale σ. Then, keypoints are computed by the difference Gaussian pyramid $\mathbf{D}(x, y, \sigma)$

$$\mathbf{D}(x, y, \sigma) = \mathbf{L}(x, y, k\sigma) - \mathbf{L}(x, y, \sigma). \tag{14}$$

SIFT utilizes the gradient information of keypoints to determine their orientation, ensuring rotation invariance of the features. Descriptor vectors of keypoints are constructed based on the gradient information within their respective neighborhoods

$$\mathbf{R}_i = \left(r_{i1}, r_{i2}, \ldots, r_{i\mathrm{m}}\right) \tag{15}$$

where R_i is the descriptor vector of keypoint i.

The gradient plays a crucial role in computing SIFT features [46], making them highly susceptible to variations in intensity. In the reflectance-corrected image, variations in reflectance characteristics among different ground objects in different bands can cause the omission of certain feature points, particularly in bands where vegetation exhibits higher reflectance (e.g., green and red edge bands), while other bands may have lower intensity. Consequently, the fused multispectral feature extraction method employs contrast-limited histogram equalization (CLAHE) to enhance individual bands of the image while minimizing the amplification of noise [47], [48]. Extraction of features from each band of the multispectral image is performed by combining image enhancement techniques with SIFT. In addition, the approach depicted in Algorithm 1 is employed to fuse the features of the multispectral image. Feature fusion is determined based on the gradient magnitude of the SIFT feature extraction, which serves as an indicator of feature significance to some extent. Feature fusion enables the extraction of band-specific features in the multispectral image, thereby providing potential feature points for subsequent 3-D reconstruction tasks.

2) Feature Matching: This step involves utilizing the descriptors obtained from feature extraction to identify and match the feature points within the image set. The feature descriptor comprises a 128-D vector constructed from gradient values in the neighborhood of the feature point, and similarity of features is assessed using the Euclidean distance

$$d(\mathbf{R}_i, \mathbf{S}_i) = \sqrt{\sum_{j=1}^{\mathrm{m}} (r_{ij} - s_{ij})^2} \tag{16}$$

where $\mathbf{S}_i$ represents the descriptor vector of the feature point i in the target image and m is the dimension of the feature descriptor.

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Algorithm 1 Overall Procedure for the Proposed Method

1: **Input**: Multi-spectral image $\mathbf{I}_k(k = 1, 2, \ldots, n)$.
2: Image enhancement for each band of $\mathbf{I}_k$.

$$\hat{\mathbf{I}}_k = \mathbf{CLAHE}(\mathbf{I}_k)$$

3: **Extraction:** Extract multispectral features F_k by SIFT operator and obtain the feature intensity:

$$\mathbf{F}_k = \mathbf{SIFT}(\hat{\mathbf{I}}_k)$$
$$\mathbf{F}_k(i) = (x, y, scale, g, orien)$$

4: **Fusion:** Fusion judgment using intensity and direction of features: For each pixel (x, y) in the image:

$$if\ \mathbf{num}(\mathbf{F}_k)in(x, y) = 1;$$
$$add\ \mathbf{F}_k\ in\ \mathbf{F}$$
$$else\ \mathbf{compared}\ \mathbf{F}_k(g)\ in\ (x, y)$$
$$add\ \mathbf{max}(\mathbf{F}_k(g))\ in\ \mathbf{F}$$

7: **Output:** fusion multi-feature $\mathbf{F}$.

To ensure accurate matching, a matching threshold is set to remove points with similarities below the threshold

$$\frac{||\mathbf{R}_i - \mathbf{S}_j||_2}{||\mathbf{R}_i - \mathbf{S}_g||_2} < \text{threshold} \tag{17}$$

where $\mathbf{S}_j$ and $\mathbf{S}_g$ represent the descriptors of the feature points in the target image that are most similar and second most similar, respectively, to the feature point in the source image.

FLANN is commonly employed for accelerated matching in the matching stage, while RANSAC is utilized for filtering out mismatched pairs. However, fused multispectral feature extraction increases the number of potential features, leading to a significant number of mismatch points in feature matching and consequently yielding unsatisfactory reconstruction results. Although RANSAC effectively eliminates false matches, a stringent threshold may also exclude several valid matching pairs. Thus, we propose an NDVI-guided multispectral feature matching method, leveraging the definition of NDVI provided in the following equation:

$$\mathbf{I}_{\text{NDVI}} = \frac{\mathbf{I}_{\text{NIR}} - \mathbf{I}_R}{\mathbf{I}_{\text{NIR}} + \mathbf{I}_R} \tag{18}$$

$$\mathbf{Mask_V} = \begin{cases} 1, & \text{if } \mathbf{I_{NDVI}} > 0.2 \\ 0, & \text{if } \mathbf{I_{NDVI}} \leq 0.2. \end{cases} \tag{19}$$

NDVI is commonly employed to extract vegetation as spectral features in multispectral images. Consequently, images can be segmented into vegetated and nonvegetated areas by computing the NDVI of the multispectral images. The features can be categorized into $\mathbf{F_V}$ and $\mathbf{F_{NV}}$ based on $\mathbf{Mask_V}$. F_V and F_{NV} are matched independently to achieve more precise matching results. The flowchart of NDVI-guided feature matching is shown in Fig. 4. By employing this matching method, mismatching of features caused by nonlinear intensity variations in multispectral images can be mitigated. Moreover, certain second-nearest neighbors are retained as accurate matches through the utilization of NDVI-guided matching, as shown in Fig. 5.

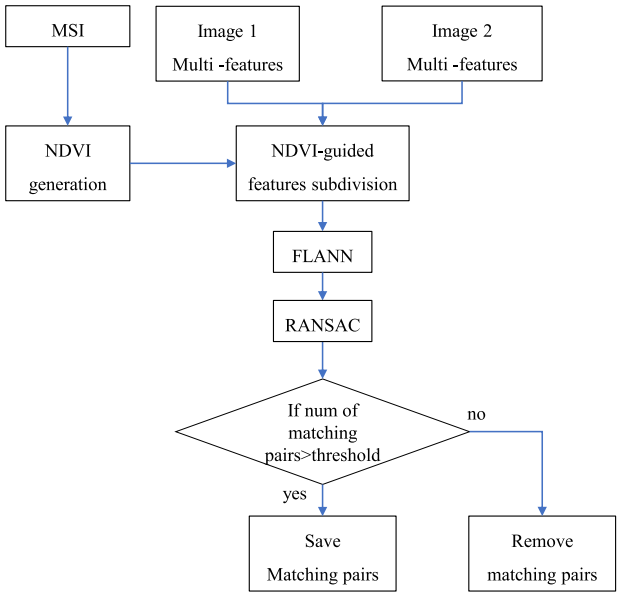


Fig. 4. Flowchart of NDVI-guided feature matching.

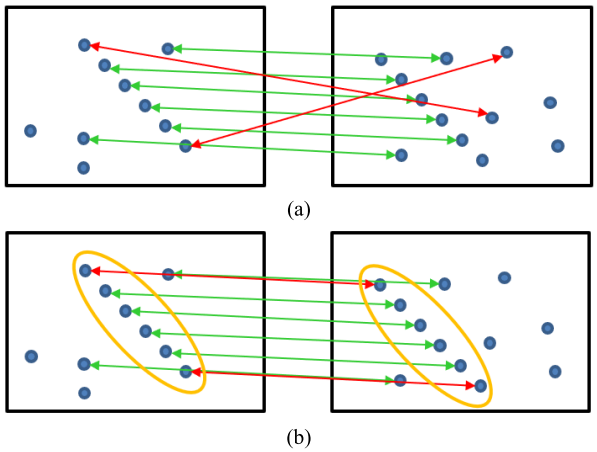


Fig. 5. Illustration of the effect of NDVI-guided feature matching. (a) Result of matching without NDVI guidance. (b) Matching result of NDVI guidance. The green line represents the correct matching pair and the red line represents the wrong matching pair. The yellow elliptical area represents the NDVI vegetation mask.

C. 3-D Information Recovery

After obtaining matched pairs of multispectral feature points, the image track can be established based on the feature matching relationship, which captures the information regarding the overlap between images [49]. During the incremental 3-D reconstruction process, the images are sequentially reconstructed using the constraints provided by the established track. By considering the coordinate position relationship of feature points in the overlapping images, the rotation and translation matrix between images can be solved using polar geometric constraints. The external parameters of the images can be determined using the perspective-n-point (PnP) method. To mitigate error accumulation during the reconstruction process and promote convergence in SFM, bundle adjustment (BA) is employed. The loss function $E(\mathbf{M}, \mathbf{X})$ for BA is

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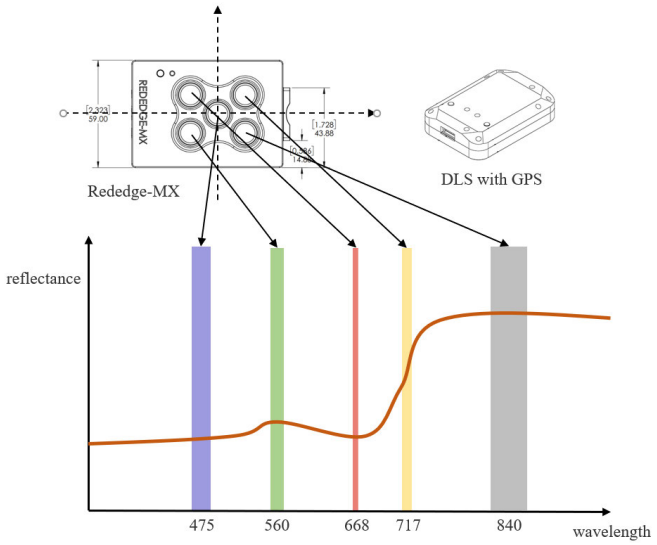


Fig. 6. Multispectral sensor RedEdge-MX and its spectral band distribution.

depicted in the following equation:

$$E(\mathbf{M}, \mathbf{X}) = \sum_{i=1}^{m} \sum_{j=1}^{n} ||x_{ij} - \mathbf{M}_i \mathbf{X}_j||_2^2 \tag{20}$$

where $\mathbf{M}_i$ is the projection matrix of camera i, $\mathbf{X}_j$ is the coordinate of reconstruction point j, and x_{ij} represents the image coordinates of feature point j in image i. The camera pose is refined to enhance the accuracy of 3-D reconstruction through the minimization of the loss function.

Using the results of sparse reconstruction as a basis, MVS technology utilizes the undistorted image and image pose information to estimate the depth map of the multispectral image [50]. This depth map, combined with the projection matrix, enables the generation of a multispectral point cloud through pixel point projection, as depicted in the following equation:

$$\mathbf{MSPC} = \left\{ X_i, Y_i, Z_i, v_i^{b1}, v_i^{b2}, \ldots, v_i^{bn} \right\} i = 1, \ldots, p_{\text{num}} \tag{21}$$

where (X_i, Y_i, Z_i) is the 3-D spatial coordinate of the ith reconstruction point, v_i^{bn} is the multispectral value of the nth band of the reconstructed point, and p_{num} represents the total number of reconstruction points.

III. Experiments and Analysis

A. Data Description

The experimental data were collected using the Rededge-MX multispectral sensor manufactured by MicaSense company.

The camera consists of five bands, and their ranges are shown in Fig. 6. The detailed parameters of the sensor are shown in Table I. Datasets of two complex scenes were used in the experiment, including Harbin Institute of Technology (HIT) campus and Zhangjiangkou Mangrove.

TABLE I
Main Parameters of the Sensor

Bands number	5 (Green, Blue, Red, Red Edge, Near Infrared)
Sensor resolution	1280 × 960 px
Weight	231.9 g
Field of view	47.2° HFOV
Bit depth	12bits
Ground sample distance at 120m	8 cm/px



(a) (b)

Fig. 7. Satellite images of the experimental scenes. (a) HIT campus. (b) Zhangjiangkou Mangrove.

1) HIT Campus: The dataset comprises 240 multispectral images captured across seven flight strips. The images overlap is 60% along the flight strips and 80% in the heading direction. The flight was conducted at a speed of 8 m/s with a spatial resolution of approximately 5 cm. The scene primarily consists of diverse vegetation, with a few buildings and vehicles present. Fig. 7(a) shows the satellite image in this dataset.

2) Zhangjiangkou Mangrove: The dataset was acquired near the Zhangjiangkou Mangrove Nature Research Center. The dataset comprises 552 multispectral images captured along ten flight strips. The images overlap is 60% along the flight path and 80% in the heading direction. The scene includes a mix of artificial buildings and mangrove vegetation. Fig. 7(b) shows the satellite image of this dataset. Standard reflectance panel and DLS data were collected simultaneously for correction.

B. Experimental Setting

The proposed 3-D reconstruction method consists of three main steps: image preprocessing, correspondence computation, and multispectral point cloud incremental reconstruction. The experimental section assesses the effectiveness of each step and demonstrates the proposed method's efficacy through visual effects and quantitative evaluation. The experiments are compared with commonly used feature extraction operators in 3-D during the feature extraction stage, and SIFT is combined with two image enhancement methods, including histogram equalization (HE) and CLAHE [42]. During the 3-D reconstruction stage, the experiments are compared with

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TABLE II
DEFINITION OF QUANTITATIVE INDICATORS FOR 3-D RECONSTRUCTION

Evaluation Indicator	Definition and Significance
Reconstructed percentage	Percentage of images involved in the reconstruction in the total images.
Feature number	Average number of features extracted from the image
Track length	The average number of images contained in the tracks of the matching relation, a larger value denotes a better geometric match
Reprojection error(m)	The error of reprojection the reconstructed points to the calibrated camera, the smaller the value, the higher the reconstruction accuracy
Reconstruction feature	The number of reconstructed feature points, larger represents more reconstruction points
Reconstruction error(m)	The spatial error between the reconstructed points and the control points collected with high-precision measuring equipment, the smaller the value indicates the higher the absolute accuracy of the reconstruction

a well-known 3-D reconstruction framework Colmap [49] and a multispectral 3-D reconstruction workflow MODM [36]. In addition, the experimental part includes a comparison with a renowned commercial software package: Pix4D. The experiment assesses multispectral 3-D reconstruction using various quantitative indicators, which are defined in Table II.

C. Experimental Results and Analysis

The 3-D reconstruction results of the two datasets are shown in Table III, and the multispectral point clouds are shown in Figs. 8 and 9. The specific analysis is given as follows.

During the reflectance correction stage of preprocessing, Fig. 10 shows that the camera's exposure time automatically adjusts due to abrupt changes in ground reflectance, leading to significant variations in the DN value. However, reflectance correction helps to achieve a more uniform intensity in the image. The calibrated intensity values are significantly more similar in the same region of two adjacent multispectral images, such as the vegetation part in the red box. The quantitative evaluation results demonstrated in Table III show that although reflectance correction reduces the number of feature extractions, it does not have an impact on the number of feature matches or the length of the track. Because the reflectivity of the image after reflectance correction is low, the number of detections for feature extraction is reduced, and more mismatches are eliminated. In the image registration stage, the multispectral image undergoes transformation and cropping, resulting in an average registration error of 0.41 pixels for the final merged multispectral image of the two datasets. This process effectively eliminates the occurrence of ghosting phenomenon, as shown in Fig. 11. The band-aligned images have sharper edges and provide accurate spectral information input for subsequent multispectral point cloud generation.

In the stage of multispectral feature extraction, we compare several commonly used feature extraction methods for 3-D reconstruction in a single baseband. The comparison reveals that the SIFT operator has superior robust performance, and the combination of image enhancement greatly enhances the number of feature extractions and matching results. The impact of reduced strength after correction is effectively mitigated in this way. Compared to HE, CLAHE changes the intensity to a lesser extent, attenuates the amplification of noise, and shows better results in the matching phase. Fused multispectral feature extraction substantially augments the number of feature extractions in an image; however, it also leads to an increase in false matches and a significant rise in reprojection error. The proposed method filters out more mismatched pairs while still retaining more feature matched pairs relative to methods such as SIFT. Due to the errors inherent in feature point extraction, there is a slight decrease in the resultant accuracy in the computation of the reprojection error. However, the cost is insignificant compared to the gain of obtaining more reconstructed points. This occurrence can be attributed to the nonlinear intensity differences observed in multispectral images. Following NDVI-guided feature matching, there is a substantial increase in the number of correct matches, resulting in a notable reduction in the reprojection error. This outcome serves as evidence for the efficacy of the multispectral feature and matching method in optimizing pose information. In Fig. 12, we distinguished vegetation areas using an NDVI threshold of 0.3. The results demonstrate a substantial presence of both correctly matched and incorrectly matched pairs after FLANN coarse matching. Following RANSAC filtering, there is a significant reduction in the number of matching pairs, effectively removing all the incorrect matches. Moreover, the NDVI-guided matching approach preserves the correct matches while further rectifying certain erroneous matches. In the results, our method has several best quantitative evaluation metrics, including track length, number of reconstructed features, and reconstruction error. Although the running time has increased, the efficiency in multiband image 3-D reconstruction tasks is still higher than that of commercial software Pix4D.

In addition, compared with the existing multispectral point cloud reconstruction method MODM, a key advantage of the proposed method is its ability to minimize spatial–spectral errors. This error is characterized by the fact that the spectral information of a given point in the multispectral point cloud does not represent its own value, but rather the spectral value of its surrounding neighboring points. Fig. 13 shows the misalignment issues between the point cloud values in single-band reconstructions due to the varying lenses used by the multispectral sensors across different bands. Furthermore, the distinct band characteristics result in a lack of one-to-one correspondence among the reconstruction points of different bands. The generation of multispectral point clouds requires employing a nearest neighbor search method. In this process, the point cloud from the reference band serves as the basis, and the adjacent point spectra from the remaining bands are assigned accordingly. The generation of multispectral point cloud requires employing a nearest neighbor search method. In this process, the point cloud from the reference band serves as the basis, and the adjacent point spectra from the remaining bands are assigned accordingly. Tables IV and V show the reconstruction results in different bands of MODM, which use the root mean square error of the nearest neighbor in the

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TABLE III
QUANTITATIVE EVALUATION RESULTS OF 3-D RECONSTRUCTION BY DIFFERENT METHODS

Method	Reconstructed percentage		Feature number		Track length		Reprojection error(m)		Reconstruction feature		time(s)		Reconstruction error(m)	
Dataset	1	2	1	2	1	2	1	2	1	2	1	2	1	2
ORB	82.1%	87.5%	4000	1302	3.61	5.41	0.66	1.17	47550	63415	**98.25**	**192.11**	0.97	0.89
HAHOG	**100%**	**100%**	4960	5258	**3.97**	**6.39**	0.41	0.46	**92825**	391173	263.84	1248.56	0.56	0.44
AKAZE	76.2%	68.5%	2174	309	3.62	5.72	0.47	0.77	29672	16970	111.93	179.69	0.72	0.65
SIFT	98.4%	**100%**	7162	8457	3.47	5.46	0.35	0.36	83652	401253	376.72	1533.57	0.57	0.44
CALI+SIFT	99.6%	**100%**	6331	6190	3.80	6.07	**0.30**	0.35	85233	429177	384.58	1646.60	0.52	0.43
HE+SIFT	**100%**	**100%**	7328	**11687**	3.73	4.84	0.35	0.34	84956	**763068**	254.54	1517.79	0.46	0.40
CLAHE+SIFT	**100%**	**100%**	**7553**	8838	3.75	5.66	**0.30**	**0.32**	89945	575500	285.47	1511.09	**0.44**	**0.36**
Colmap	80.6%	81.7%	6075	5874	3.24	5.47	0.43	**0.38**	128390	805877	1912.71	7918.43	-	-
MODM	99.6%	100%	6412	6046	3.75	5.34	0.38	0.40	89433	446576	2061.45	7769.57	0.54	0.57
MIFU	**100%**	**100%**	**24674**	**24248**	3.73	5.18	1.15	1.07	151077	822439	**1392.73**	**6235.27**	0.95	0.82
Pix4D	99%	**100%**	19379	16245	-	-	**0.34**	0.40	143602	817943	2880.27	14190.68	0.46	0.44
Ours	**100%**	**100%**	**24674**	**24248**	**4.07**	**6.72**	0.36	**0.38**	**160280**	**874677**	1482.46	6462.32	**0.42**	**0.37**

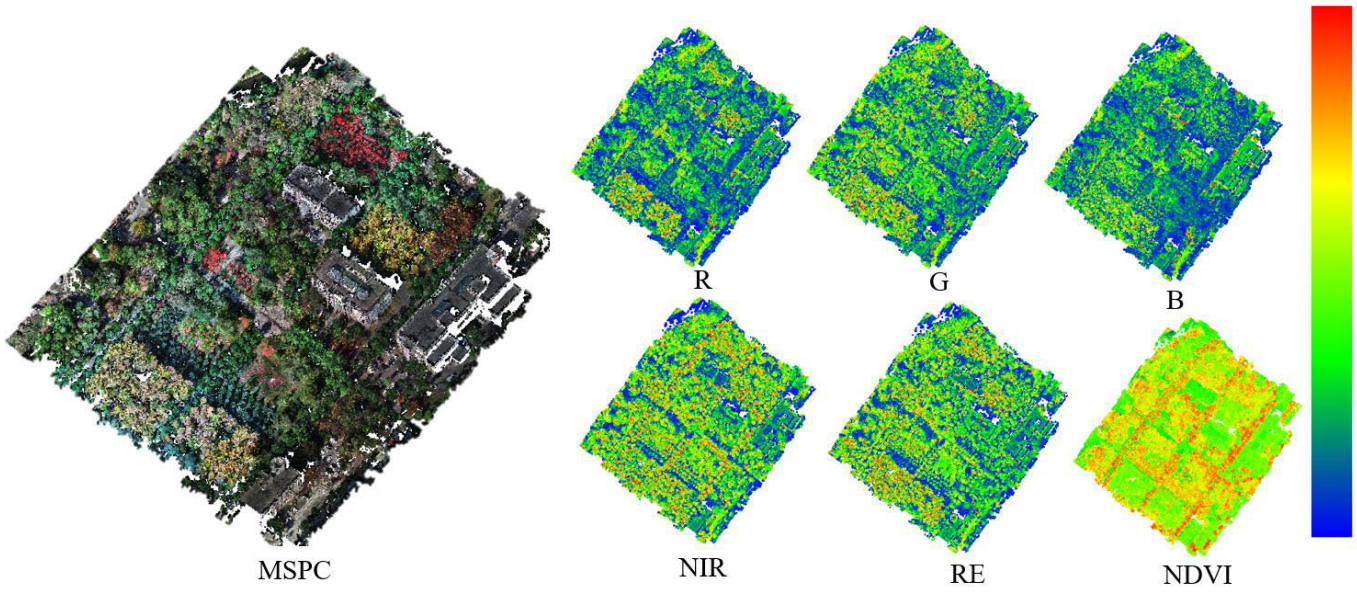


Fig. 8. Multispectral point cloud of HIT campus.

reconstruction results of different bands as a measure of the spatial spectrum error. As mentioned in [40], MODM automatically detects the presence of multiband images and presents slight divergences in reconstructing the multispectral model. Our proposed method effectively mitigates the spatial errors between different lenses and avoids mapping errors between different bands by using registered multispectral images as input to obtain multispectral point cloud. The spatial–spectral error in the reconstruction results is solely attributed to band alignment errors during the preprocessing step, which is significantly lower than MODM results. Compared with other methods, the proposed method has better reconstruction performance and higher efficiency.

To test the robustness of the proposed method against image noise, we have added two common image noises: Gaussian noise and salt and pepper noise in the HIT campus dataset. Several methods were used to compare the 3-D reconstruction tests on image sets containing simulated noise. In the test results, the number of features extracted and the length of the trajectory represent the robustness of the feature

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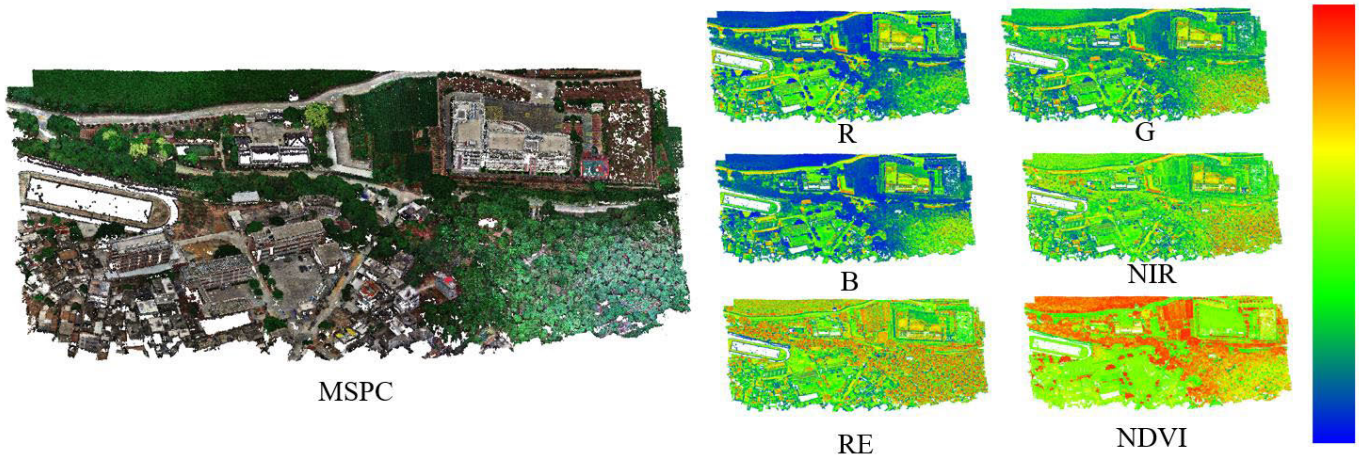


Fig. 9. Multispectral point cloud of Zhangjiangkou Mangrove.

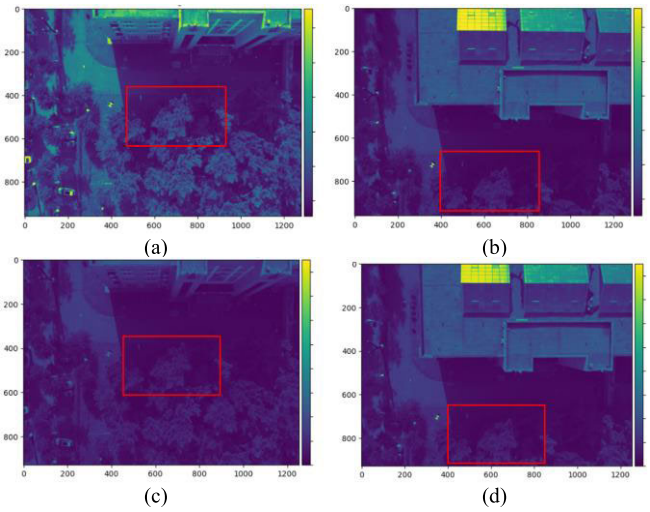


Fig. 10. Results of reflectance correction of multispectral images. (a) and (b) Red band images of adjacent multispectral images. (c) and (d) Results after reflectance correction, the red boxes correspond to local vegetation areas.

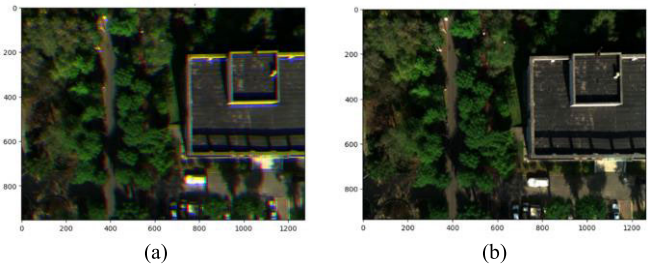


Fig. 11. Results of RGB band combination for multispectral images. (a) Before band alignment. (b) After band alignment.

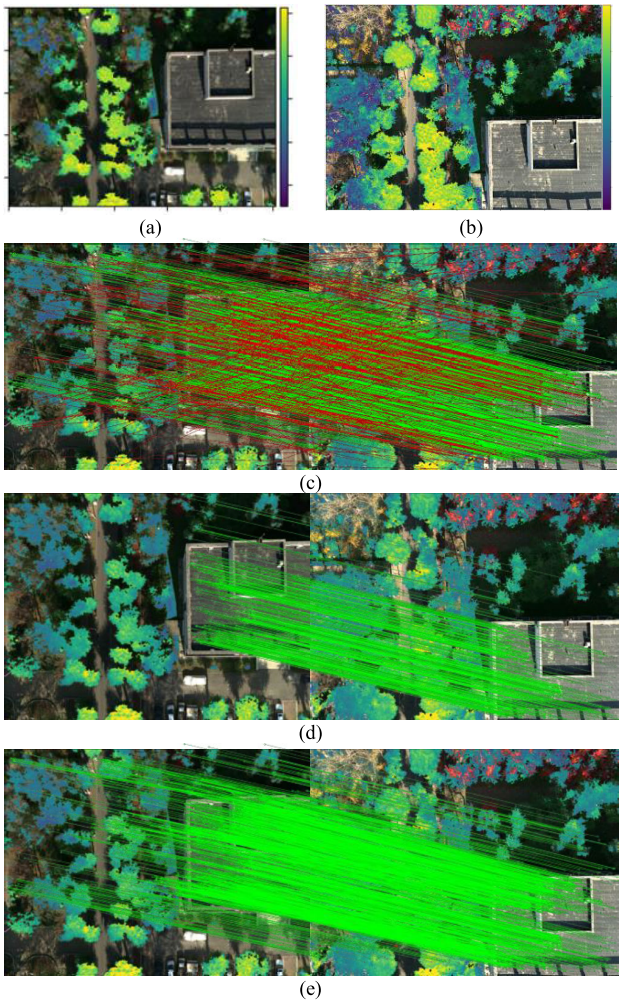
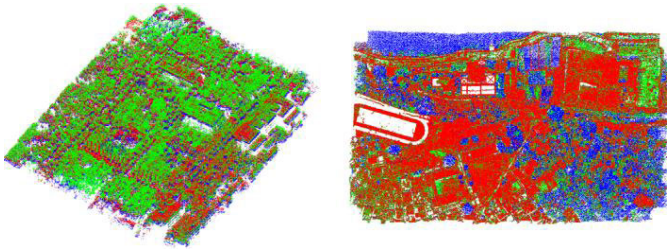


Fig. 12. NDVI masks and feature matching results. (a) and (b) NDVI masks of two adjacent multispectral images. Vegetation areas are rendered using NDVI values and overlaid on the RGB band image for display. (c) FLANN coarse matching result. (d) RANSAC matching result. (e) NDVI-guided matching results. Red lines represent incorrect matches and green lines are correct matches.

extraction method to noise. The number of reconstruction points and the reprojection error represent the quality of the 3-D reconstruction. The results in Table VI show that the commonly used 3-D reconstruction feature extraction methods exhibit a significant decrease in reconstruction accuracy under the influence of Gaussian noise and salt and pepper noise. Especially in salt and pepper noise, it has a notable impact on intensity-dependent feature extraction operators. The proposed method benefits from the filtering effect of the NDVI mask

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(a) (b)

Fig. 13. Three single-band (red, green, and blue) reconstructions are superimposed to show the spatial error between point clouds in different bands, with different color dots representing the reconstruction results of different bands. (a) HIT campus. (b) Zhangjiangkou Mangrove.

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF HIT CAMPUS IN DIFFERENT WAVEBANDS BY MODM

Spectral band	Rec-percentage	Feature number	Reprojection error(m)	Rec-feature	Nearest error
band1	99.6%	6207	0.29	100846	0.58
band2	99.6%	6331	0.30	85233	0(base)
band3	100%	6170	0.33	98966	0.45
band4	99.2%	6653	0.40	78060	0.52
band5	99.2%	6505	0.38	80038	0.34

TABLE V
QUANTITATIVE EVALUATION RESULTS OF ZHANGJIANGKOU MANGROVE IN DIFFERENT WAVEBANDS BY MODM

Spectral band	Rec-percentage	Feature number	Reprojection error(m)	Rec-feature	Nearest error
band1	100%	5433	0.37	356009	0.41
band2	100%	6190	0.35	429177	0(base)
band3	100%	5722	0.38	376849	0.24
band4	100%	6331	0.38	526511	0.49
band5	100%	6188	0.37	490150	0.16

TABLE VI
QUANTITATIVE EVALUATION RESULT IN DIFFERENT NOISE OF HIT

Noise	Method	Feature number	Repro-error(m)	Track length	Rec-feature
Gaussian	ORB	1362	0.86	3.41	39874
	SIFT	5426	0.73	3.61	81286
	Our	**23186**	**0.62**	**3.92**	**147824**
Salt and pepper	ORB	1486	-	-	-
	SIFT	5618	0.66	3.58	56424
	Our	**22498**	**0.63**	**3.84**	**129266**

to obtain better reconstruction results under the influence of noise.

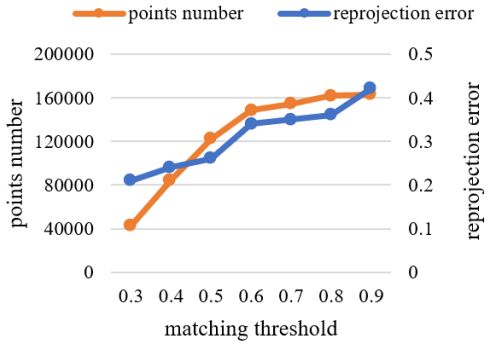
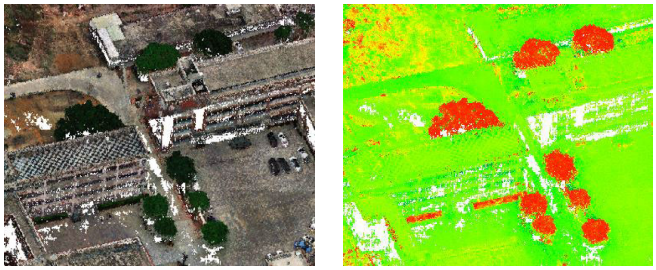


Fig. 14. Effect of the matching threshold parameter.



(a) (b)

Fig. 15. Local multispectral point cloud and NDVI rendering results. (a) Multispectral point cloud. (b) NDVI display of Zhangjiangkou Mangrove.

The experiments also tested the parameter sensitivity of the matching threshold. The matching threshold influences the quantity of feature matching pairs. A lower matching threshold results in higher accuracy of retained matching pairs and reduced reprojection error; nevertheless, it also decreases the number of reconstructed points. Fig. 14 shows that when the matching threshold falls below 0.6, a lower reprojection error is achievable, but this comes at the cost of a significantly reduced number of reconstructed points compared to higher matching thresholds. Conversely, with a matching threshold exceeding 0.8, the expense of boosting reconstruction points outweighs the loss associated with an increase in reprojection error. In our experimental method, we set the matching threshold to 0.8 for balanced results.

The accurate 3-D spatial information and multispectral data enable the extension of certain vegetation indices into 3-D space for calculation. Fig. 15 shows the excellent ability of the NDVI point cloud to accurately distinguish vegetation in 3-D, effectively replacing the need for complex vegetation segmentation methods. This highlights the potential of multispectral point cloud applications in tasks such as vegetation classification.

IV. CONCLUSION

This article presents a novel method for generating multispectral point cloud based on 3-D reconstruction of multispectral images. The proposed method exhibits improved robustness, higher efficiency, and reduced spatial–spectral error compared to existing 3-D reconstruction methods. According to the characteristics of multispectral sensors and multispectral images, image preprocessing is employed

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to mitigate intensity differences and spatial errors between images. The feature extraction and matching stage of the 3-D reconstruction process incorporates a multispectral fused feature extraction algorithm and an NDVI-guided feature matching algorithm, facilitating the extraction of a larger number of feature points and matching pairs. Ultimately, by employing incremental reconstruction and spectral mapping techniques, the proposed method achieves high-precision multispectral point cloud data.

Through the experimental verification of the two datasets and the analysis of the experimental results, it can be concluded that the proposed method has made targeted improvements to the multispectral point cloud method based on the 3-D reconstruction of multispectral images and has achieved significant improvement. The proposed method efficiently and accurately generates multispectral point cloud data, which offers valuable data support for remote sensing applications.

REFERENCES

[1] Q. Wang and Y. Gu, "A discriminative tensor representation model for feature extraction and classification of multispectral LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 1568–1586, Mar. 2020.

[2] N. Ekhtari, C. Glennie, and J. C. Fernandez-Diaz, "Classification of airborne multispectral LiDAR point clouds for land cover mapping," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 6, pp. 2068–2078, Jun. 2018.

[3] J. Xia, N. Yokoya, and A. Iwasaki, "Fusion of hyperspectral and LiDAR data with a novel ensemble classifier," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 6, pp. 957–961, Jun. 2018.

[4] L. Chen, Y. Gu, X. Li, X. Zhang, and B. Liu, "A normalized spatial–spectral supervoxel segmentation method for multispectral point cloud data," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5704311.

[5] O. M. Saad, S. Fomel, R. Abma, and Y. Chen, "Unsupervised deep learning for 3D interpolation of highly incomplete data," *Geophysics*, vol. 88, no. 1, pp. WA189–WA200, Jan. 2023.

[6] W. Huang, D. Feng, and Y. Chen, "De-aliased and de-noise Cadzow filtering for seismic data reconstruction," *Geophys. Prospecting*, vol. 68, no. 2, pp. 553–571, Feb. 2020.

[7] Y. Chen, M. Bai, and Y. Chen, "Obtaining free USArray data by multi-dimensional seismic reconstruction," *Nature Commun.*, vol. 10, no. 1, pp. 1–10, Sep. 2019.

[8] K. Zhang, M. Wang, S. Yang, and L. Jiao, "Convolution structure sparse coding for fusion of panchromatic and multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1117–1130, Feb. 2019.

[9] Z. Qin, X. Li, and Y. Gu, "An illumination estimation and compensation method for radiometric correction of UAV multispectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5545012.

[10] T. Li, T. Liu, Y. Wang, X. Li, and Y. Gu, "Spectral reconstruction network from multispectral images to hyperspectral images: A multitemporal case," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5535016.

[11] D. Hong, J. Kang, N. Yokoya, and J. Chanussot, "Graph-induced aligned learning on subspaces for hyperspectral and multispectral data," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4407–4418, May 2021.

[12] L. Gao, D. Hong, J. Yao, B. Zhang, P. Gamba, and J. Chanussot, "Spectral superresolution of multispectral imagery with joint sparse and low-rank learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2269–2280, Mar. 2021.

[13] S. Luo et al., "Comparative performances of airborne LiDAR height and intensity data for leaf area index estimation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 1, pp. 300–310, Jan. 2018.

[14] X. Ji, B. Yang, Y. Wang, Q. Tang, and W. Xu, "Full-waveform classification and segmentation-based signal detection of single-wavelength bathymetric LiDAR," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4208714.

[15] Z. Zhang, T. Li, X. Tang, X. Lei, and Y. Peng, "Introducing improved transformer to land cover classification using multispectral LiDAR point clouds," *Remote Sens.*, vol. 14, no. 15, p. 3808, Aug. 2022.

[16] Q. Wang, Y. Gu, M. Yang, and C. Wang, "Multi-attribute smooth graph convolutional network for multispectral points classification," *Sci. China Technol. Sci.*, vol. 64, no. 11, pp. 2509–2522, Nov. 2021.

[17] Z. Jing et al., "Multispectral LiDAR point cloud classification using SE-PointNet++," *Remote Sens.*, vol. 13, no. 13, p. 2516, Jun. 2021.

[18] J. A. R. Rall and R. G. Knox, "Spectral ratio biospheric LiDAR," *IEEE Int. Geosci. Remote Sens. Symp.*, vol. 3. Sep. 2004, pp. 1951–1954.

[19] G. Mandlburger, M. Pfennigbauer, F. Steinbacher, and N. Pfeifer, "Airborne hydrographic LiDAR mapping—Potential of a new technique for capturing shallow water bodies," in *Proc. 19th Int. Congr. Modeling Simulation*, Dec. 2011, pp. 2416–2422.

[20] J. Fernandez-Diaz et al., "Capability assessment and performance metrics for the Titan multispectral mapping LiDAR," *Remote Sens.*, vol. 8, no. 11, p. 936, Nov. 2016.

[21] S. Kaasalainen, T. Lindroos, and J. Hyyppa, "Toward hyperspectral LiDAR: Measurement of spectral backscatter intensity with a supercontinuum laser source," *IEEE Geosci. Remote Sens. Lett.*, vol. 4, no. 2, pp. 211–215, Apr. 2007.

[22] W. Li et al., "A liquid crystal tunable filter-based hyperspectral LiDAR system and its application on vegetation red edge detection," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 291–295, Feb. 2019.

[23] Z. Niu et al., "Design of a new multispectral waveform LiDAR instrument to monitor vegetation," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1506–1510, Jul. 2015.

[24] G. Sun et al., "32-channel hyperspectral waveform LiDAR instrument to monitor vegetation: Design and initial performance trials," *Proc. SPIE*, vol. 9263, pp. 533–539, Nov. 2014.

[25] P. Ghamisi et al., "Multisource and multitemporal data fusion in remote sensing: A comprehensive review of the state of the art," *IEEE Geosci. Remote Sens. Mag.*, vol. 7, no. 1, pp. 6–39, Mar. 2019.

[26] J. U. H. Eitel et al., "Beyond 3-D: The new spectrum of LiDAR applications for earth and ecological sciences," *Remote Sens. Environ.*, vol. 186, pp. 372–392, Dec. 2016.

[27] K. Koenig and B. Höfle, "Full-waveform airborne laser scanning in vegetation studies—A review of point cloud and waveform features for tree species classification," *Forests*, vol. 7, no. 12, p. 198, Sep. 2016.

[28] T. Zhou, S. M. Hasheminasab, and A. Habib, "Tightly-coupled camera/LiDAR integration for point cloud generation from GNSS/INS-assisted UAV mapping systems," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 336–356, Oct. 2021.

[29] L. Zhou, Z. Li, and M. Kaess, "Automatic extrinsic calibration of a camera and a 3D LiDAR using line and plane correspondences," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Oct. 2018, pp. 5562–5569.

[30] Y. Gu, C. Wang, and X. Li, "An intensity-independent stereo registration method of push-broom hyperspectral scanner and LiDAR on UAV platforms," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5540014.

[31] M. Brell, C. Rogass, K. Segl, B. Bookhagen, and L. Guanter, "Improving sensor fusion: A parametric method for the geometric coalignment of airborne hyperspectral and LiDAR data," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3460–3474, Jun. 2016.

[32] Y. Gu, X. Jin, R. Xiang, Q. Wang, C. Wang, and S. Yang, "UAV-based integrated multispectral-LiDAR imaging system and data processing," *Sci. China Technol. Sci.*, vol. 63, no. 7, pp. 1293–1301, Jul. 2020.

[33] A. Lucieer, S. M. D. Jong, and D. Turner, "Mapping landslide displacements using structure from motion (SfM) and image correlation of multi-temporal UAV photography," *Prog. Phys. Geography, Earth Environ.*, vol. 38, no. 1, pp. 97–116, Feb. 2014.

[34] O. Nevalainen et al., "Individual tree detection and classification with UAV-based photogrammetric point clouds and hyperspectral imaging," *Remote Sens.*, vol. 9, no. 3, p. 185, Feb. 2017.

[35] J. A. Gonçalves and R. Henriques, "UAV photogrammetry for topographic monitoring of coastal areas," *ISPRS J. Photogramm. Remote Sens.*, vol. 104, pp. 101–111, Jun. 2015.

[36] S. Jiang, C. Jiang, and W. Jiang, "Efficient structure from motion for large-scale UAV images: A review and a comparison of SfM tools," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 230–251, Sep. 2020.

[37] O. Ozyesil, V. Voroninski, R. Basri, and A. Singer, "A survey of structure from motion," *Acta Numerica*, vol. 26, pp. 305–364, May 2017.

[38] R. Gomez-Ojeda, F.-A. Moreno, D. Zuñiga-Noël, D. Scaramuzza, and J. Gonzalez-Jimenez, "PL-SLAM: A stereo SLAM system through the combination of points and line segments," *IEEE Trans. Robot.*, vol. 35, no. 3, pp. 734–746, Jun. 2019.

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[39] K. Ito, T. Ito, and T. Aoki, "PM-MVS: PatchMatch multi-view stereo," *Mach. Vis. Appl.*, vol. 34, no. 2, pp. 1–12, Mar. 2023.
[40] A. Vong et al., "How to build a 2D and 3D aerial multispectral map?—All steps deeply explained," *Remote Sens.*, vol. 13, no. 16, p. 3227, Aug. 2021.
[41] M. Bansal, M. Kumar, and M. Kumar, "2D object recognition: A comparative analysis of SIFT, SURF and ORB feature descriptors," *Multimedia Tools Appl.*, vol. 80, no. 12, pp. 18839–18857, May 2021.
[42] W. Ma et al., "Remote sensing image registration with modified SIFT and enhanced feature matching," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 1, pp. 3–7, Jan. 2017.
[43] R. Feng, H. Shen, J. Bai, and X. Li, "Advances and opportunities in remote sensing image geometric registration: A systematic review of state-of-the-art approaches and future research directions," *IEEE Geosci. Remote Sens. Mag.*, vol. 9, no. 4, pp. 120–142, Dec. 2021.
[44] X. Liu, X. Chen, M. Bai, and Y. Chen, "Time-lapse image registration by high-resolution time-shift scan," *Geophysics*, vol. 86, no. 3, pp. M49–M58, May 2021.
[45] S. Fomel and L. Jin, "Time-lapse image registration using the local similarity attribute," *Geophysics*, vol. 74, no. 2, pp. A7–A11, Mar. 2009.
[46] S.-J. Chen, H.-L. Shen, C. Li, and J. H. Xin, "Normalized total gradient: A new measure for multispectral image registration," *IEEE Trans. Image Process.*, vol. 27, no. 3, pp. 1297–1310, Mar. 2018.
[47] G. R. Vidhya and H. Ramesh, "Effectiveness of contrast limited adaptive histogram equalization technique on multispectral satellite imagery," in *Proc. Int. Conf. Video Image Process.*, Dec. 2017, pp. 234–239.
[48] Y. Zhang and T. Wang, "Multispectral image registration based on an improved scale-invariant feature transform algorithm," *J. Appl. Remote Sens.*, vol. 16, no. 2, May 2022, Art. no. 024515.
[49] J. L. Schönberger and J.-M. Frahm, "Structure-from-motion revisited," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 4104–4113.
[50] J. L. Schonberger, E. L. Zheng, J. M. Frahm, and M. Pollefeys, "Pixelwise view selection for unstructured multi-view stereo," in *Proc. Eur. Conf. Comput. Vis.*, in Lecture Notes in Computer Science, 2016, pp. 501–518.



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